# Alexandre BELLEMIN-MAGNINOT . Jarvis Consulting

With a double degree in Information Technology from ÉTS Montréal and an engineering degree specializing in computer science from the Institut National des Sciences Appliquées de Lyon (INSA Lyon, France), I have a hybrid profile combining data engineering, decision analytics, and project coordination. Throughout my career, I have implemented ETL pipelines in medallion architecture with Azure Data Factory and SQL procedures, extracted data via ERP APIs, processed data with Databricks, and delivered Power BI dashboards for analysis. I have strengthened my communication, process optimization, and cross-functional collaboration skills by delivering automation tools that have improved data reliability and reduced the execution time of certain processes from a week to a few minutes. My goal is to solve data-related problems, improve processes, and create impactful data solutions to develop my skills.

## Skills

**Proficient:** Python (Pandas/NumPy), Power BI, PySpark, Databricks, Microsoft Azure, SGBD/SQL, Excel/VBA, Java, Linux/Bash, Docker, Agile/Scrum, Git, MS Power Automate

**Competent:** Django/Flask, R, C/C++, PHP / Laravel, MS Project

**Familiar:** MongoDB, Tableau, Javascript, Unity, Matlab

## Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_AlexandreBellemin-Magninot

**Databricks ETL Medallion & Delta Live Table Pipelines** [GitHub]: Designed and implemented end-to-end ETL pipelines in Databricks following the Medallion Architecture (Bronze/Silver/Gold) to process millions of transactional records with fraud indicators. Built modular notebooks and multiple ingestion strategies, delivering analytics-ready datasets and dashboards. Developed a Delta Live Tables (DLT) pipeline for stock market data ingestion with enforced data quality and reliable transformation workflows.

**PySpark Customer Analytics Migration.** [GitHub]: Migrated a Pandas-based retail analytics workflow to PySpark on Databricks (Spark 3.5.2), redesigning transformations using the structured DataFrame API for scalable distributed processing. Implemented customer segmentation (RFM), KPI computation, and optimized queries without Python UDFs to ensure performance on large datasets. Deployed and tested workflows on a multi-node Spark cluster (Google Dataproc).

**Retail Analytics & Customer Segmentation PoC.** [GitHub]: Designed and implemented a complete analytics workflow for a simulated e-commerce company to help understand customer behavior and revenue sources. Built a reproducible environment using Docker, PostgreSQL, and Jupyter notebook. Performed data cleaning and transformations, exploratory analysis, KPI monitoring (monthly sales revenue/growth, number of new and existing customers…), and RFM segmentation using Python, Pandas, NumPy, and Matplotlib. Identified key customer segments, retention risks, and growth opportunities. This project strengthened my data engineering foundations (pipelines, ETL, database setup) and my ability to translate raw data into actionable business insights.

**Java Grep Application** [GitHub]: Developed a command-line Java application that replicates core Linux `grep` functionality by recursively scanning directories and matching file contents using regular expressions. Implemented two distinct approaches: a traditional file system-based solution and a functional implementation using Java Streams and lambda expressions. Built with Core Java, Maven, and SLF4J logging, version-controlled with Git, and containerized with Docker to ensure portability and reproducible execution. This project strengthened my understanding of Java I/O, functional programming, and application packaging.

**Linux Cluster Monitor.** [GitHub]: Developed a lightweight Linux monitoring solution that collects hardware specifications and real-time system metrics across multiple distributed servers. Built using Bash for data collection, PostgreSQL containerized with Docker as RDBMS, and cron jobs for automated ingestion. The project demonstrates how to design a simple DevOps-style data pipeline for capacity planning and system observability.

## Highlighted Projects

**AGILE Optimization Project (INSA Lyon).**: Designed a route-optimization system for a simulated delivery service using Dijkstra's algorithm to compute the shortest and most efficient paths between warehouses and delivery points. Developed in Java within an Agile/Scrum framework (sprints, backlog, Git-based collaboration). Strengthened skills in algorithmic problem-solving, team coordination, and Agile project execution.

**C Compiler (INSA Lyon).**: Built a simplified C compiler using C++ and assembly to understand compilation stages (lexing, parsing, code generation). Applied low-level programming, language theory, and modular software design. Used GitHub for version control, issues, and team collaboration.

**Connected Doorbell (ÉTS Montréal).**: Developed a Flask web application deployed on Azure Web App to support a connected doorbell system. Implemented real-time notifications and automated image capture stored in Azure Blob Storage. Explored IoT-style communication, cloud deployment, and backend development.

## Professional Experiences

**Software Developer (data engineering)., Jarvis (2025-present)**: - Developed data engineering skills by working on end-to-end projects involving SQL, PySpark (Databricks), Python (Pandas/NumPy), Java (Stream API), Bash, Docker, and Git, collaborating in an Agile environment. - Strengthened data engineering fundamentals, deployed containerized applications, and worked in sprints.

**Data Engineer / Project Coordinator., Bombardier (09/2024 - 05/2025)**: - Implementation of a simplified ETL pipeline for a cross-functional team, enabling the retrieval of sensitive data distributed across several thousand company employees and making it usable for a dozen different projects. - Designed and developed automated tools with Microsoft 365 (Excel/VBA, Power Automate, Planner, SharePoint) to increase the reliability of the data used and optimize internal processes, enabling some to reduce their execution time from one week to 5 minutes. - Creation, maintenance, and presentation of dashboards to governance using Power BI. Sourced from Excel on SharePoint, and use of DAX queries to create metrics. - Supported the training of hundreds of new engineers, including planning, session monitoring, logistical assistance, and provision of educational resources. - Contributed to knowledge management for several thousand employees by structuring information flows and facilitating governance. - Participated in the development of internal podcasts designed to archive and disseminate tacit knowledge. - Communicated primarily in English.

**Data Consultant., Thélio (05/2023 - 08/2023)**: - Strengthening skills in ETL/ELT, data modeling, cloud integration, and analytics. - Implementation of ETL pipelines in medallion architecture using SQL stored procedures, from ingestion to visualization on Power BI. Automation of these pipelines via Azure Data Factory. - Design and development of an ERP module with Laravel (PHP) for managing company offers and implementation of automated extraction of this data to a data warehouse via API calls. - Implementation of a medallion architecture for company CSR data with SQL. - Implementation of a CSR dashboard in Power BI for monitoring environmental footprint indicators. - Analysis and processing of data as part of a project conducted with Olympique Lyonnais, using Databricks.

**Web developer., Ancey Informatique. (07/2022 - 08/2022)**: - Developed automated tests in PHPUnit for a Laravel resource management platform used by several clients. - Worked on database modeling, query optimization, and maintenance of open-source tools. Gained experience in backend development and testing methodologies.

## Education

**École de Technologie Supérieure (ÉTS Montréal) (2023-2025)**, Master of Information Technology (IT)., Information Technology. - Courses taken: Business Intelligence (BI), Cloud computing, Virtual Reality (VR), Knowledge and innovation management, Project portfolio management, Feasibility analysis, AGILE, Artificial intelligence management. - GPA: 3.7/4.0

**Institut National des Sciences Appliquées (INSA Lyon, FRANCE) (2019-2025)**, Master's in Engineering., Computer Science. - Courses taken: Software development and design, DBMS, OLAP data modeling, Networks, Machine Learning, AGILE, Data mining, Statistics, Probability. - Animation manager at Student Association of the Computer Science Department. - Consultant for the school's junior company ETIC INSA Technologies: understanding the customer's needs and implementing the solution. - GPA: 3.7/4.0

## Miscellaneous

- Drummer in several bands, playing styles ranging from pop to metal.
- Mountain sports enthusiast: hiking, skiing, and climbing.
- Amateur cook, with a particular interest in creating sauces.
- Interest in the physics behind space phenomena and astrophysical concepts.